Group: Phuc & Hieu

CHINOOK SALES DATA ANALYSIS

Business Data Management

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TABLE OF CONTENTS

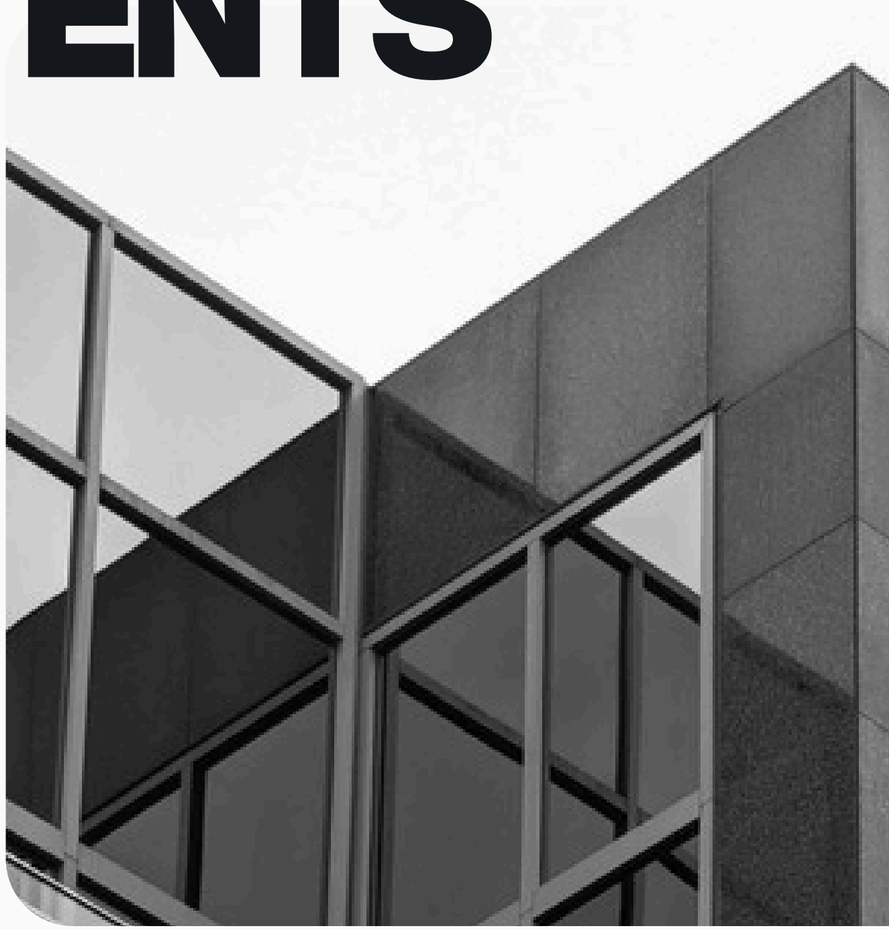
1. Introduction & Business Problem

2. Methodology & Database Overview

3. Data Cleansing & Transformation

4. Data Exploration & Insights

5. Strategic Recommendations



THE BUSINESS PROBLEM

Chinook is a global digital media store.
Experiencing fluctuating yearly revenue.
Need to identify: Most **profitable markets**, **top music genres**, and overall sales trends **to optimize business strategies.**

METHODOLOGY & DATABASE OVERVIEW

Database Overview (The Chinook Dataset)

Context: A relational database simulating a global B2C digital media store (**similar to iTunes**).

Structure: 11 interconnected tables structured via Primary/Foreign Keys.

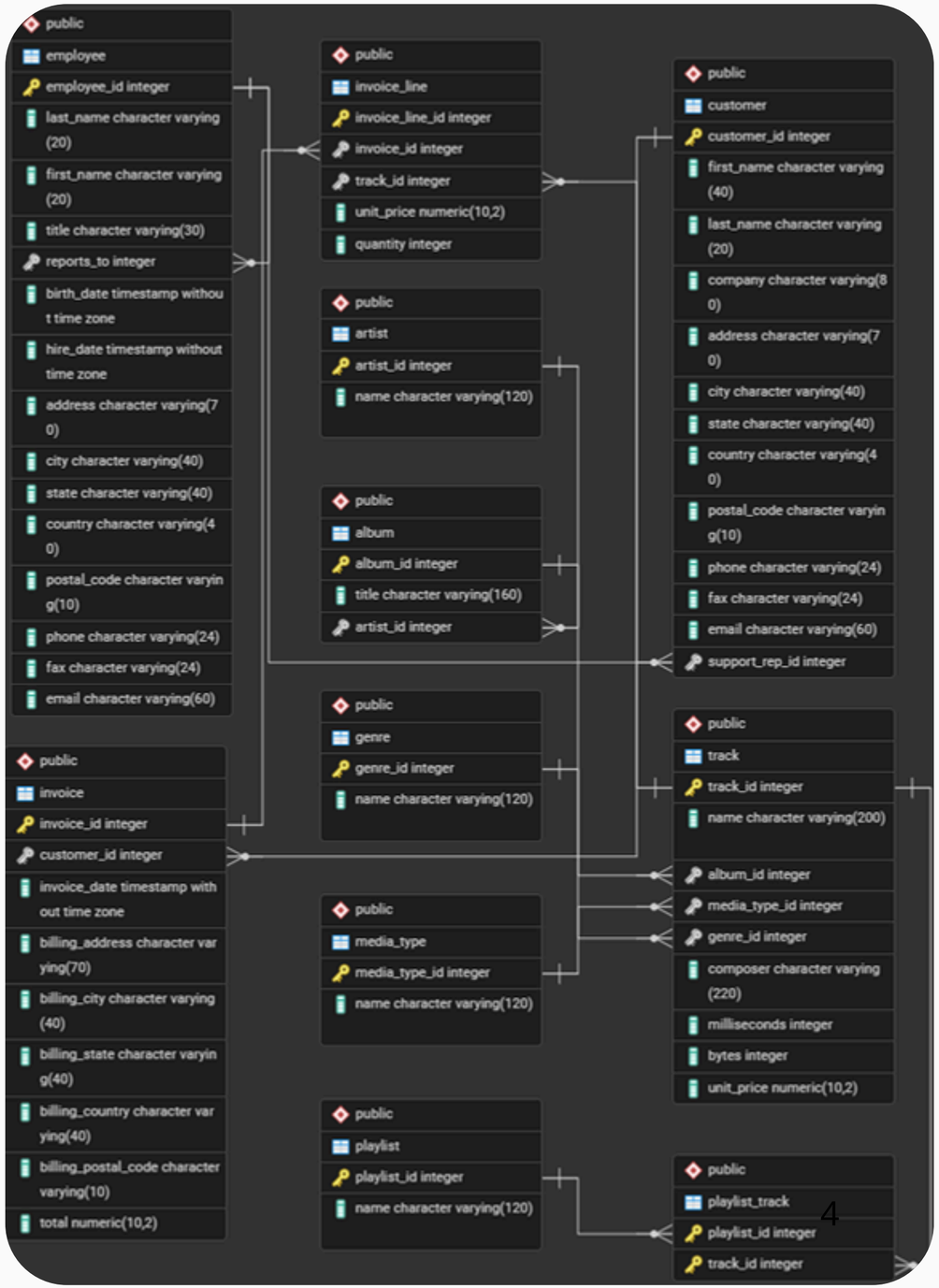
Core Business Entities:

Sales/Transactions: invoice, invoice_line (Tracking revenue).

Product Catalog: track, genre, album (Digital inventory).

CRM & HR: customer, employee (Client profiles & Sales agents).

PostgreSQL → jupyter



DATA CLEANSING & TRANSFORMATION

- **Goal:** Transform raw transactional data into a structured, business-ready format.
- **1. Temporal Variable Extraction:**
  - **SQL:** Used EXTRACT(YEAR FROM invoice_date) to isolate the year for time-series tracking.
  - **Python:** Applied .astype(int) to remove floating decimals (e.g., converting 2021.0 →→2021 for clean axis labels).
- **2. Metric Aggregation & Noise Reduction:**
  - **SQL:** Summarized massive line items using SUM() and GROUP BY dimensions (Country, Year, Genre).

Q1. GEOGRAPHIC REVENUE ANALYSIS

- **Mindset**: Where is our cash flow primarily coming from?
- **Data Direction:**
  - **Table**: invoice
  - **Target Columns**: billing_country & total
- **Execution (SQL):**
  - Aggregated data using SUM(total).
  - Categorized by country using GROUP BY.
  - Ranked top markets using ORDER BY ... DESC.

```
query = """
SELECT billing_country, SUM(total) AS revenue
FROM invoice
GROUP BY billing_country
ORDER BY revenue DESC
"""

df_country = pd.read_sql_query(query, engine)
df_country.head()
```

	billing_country	revenue
0	USA	523.06
1	Canada	303.96
2	France	195.10
3	Brazil	190.10
4	Germany	156.48

INSIGHT 1

MARKET EXPANSION STRATEGY

Outcome:

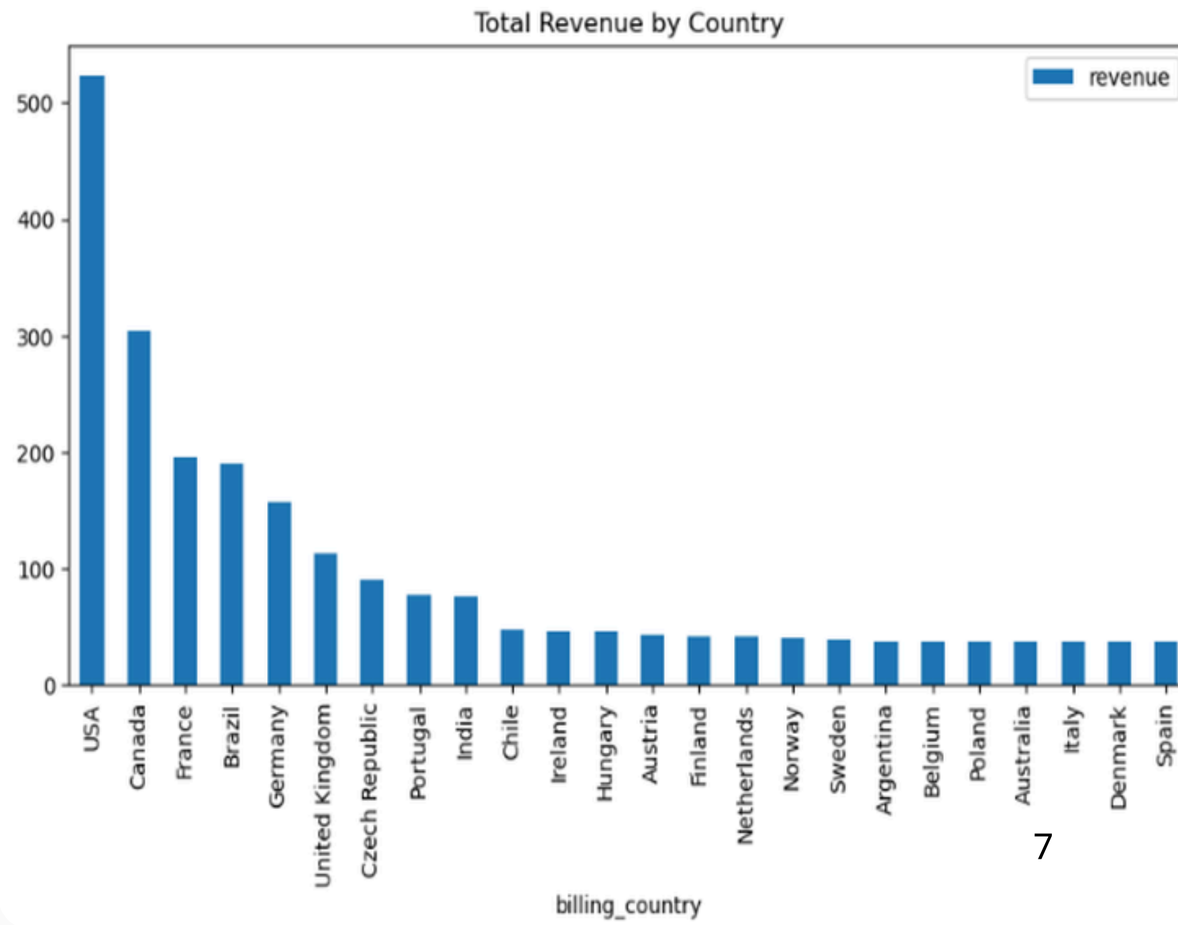
- USA absolutely dominates the market ($523.06).
- Followed by Canada (303.96) and France (195.10).

Business Implication:

- North America is Chinook's "Cash Cow" (core profit center).

Action: Marketing budgets and VIP customer retention strategies must be heavily concentrated in the USA and Canada to maximize ROI.

```python
import matplotlib.pyplot as plt

df_country.plot(
    x="billing_country",
    y="revenue",
    kind="bar",
    figsize=(10,5),
    title="Total Revenue by Country"
)

plt.show()
```



Q2. YEAR-OVER-YEAR

SALES TREND

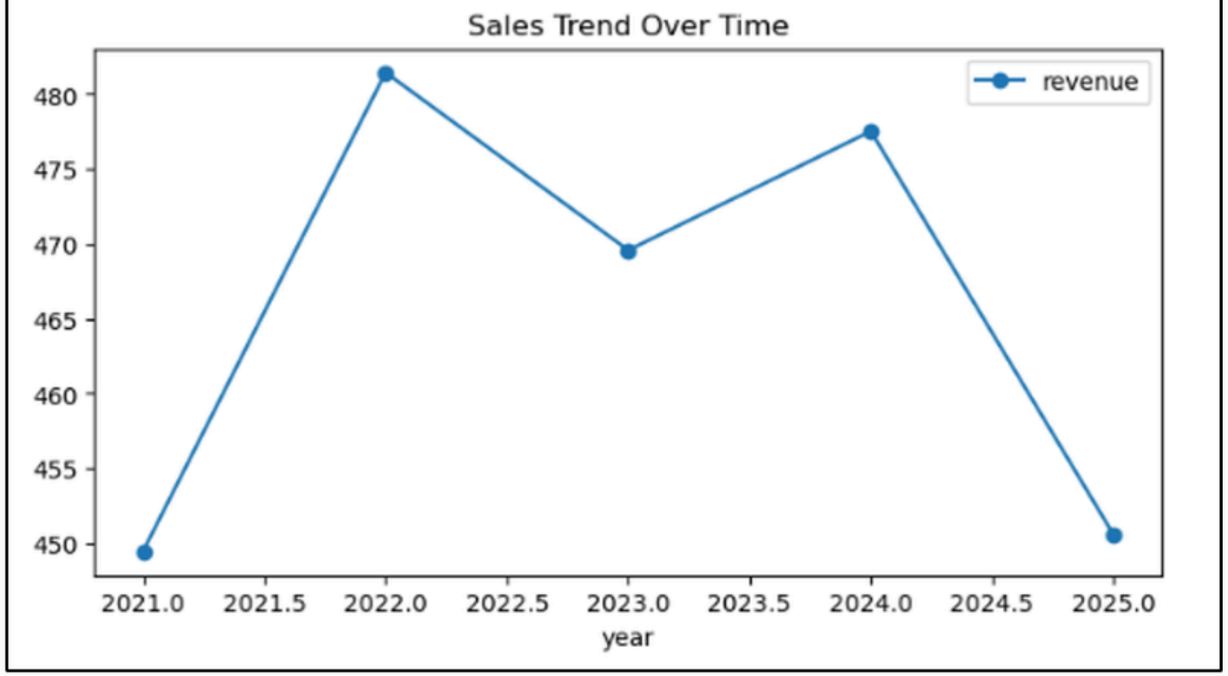
- **Mindset**: Is the company's financial health growing steadily or declining?
- **Data Direction:**
  - **Table:** invoice
  - **Target Column:** invoice_date (Raw data is in TIMESTAMP format).
- **Execution (SQL & Python):**
  - **SQL:** Used EXTRACT(YEAR FROM...) to isolate the exact year.
  - **SQL:** Used GROUP BY year and SUM() for total revenue.
  - **Python:** Cleaned x-axis data using .astype(int) to remove decimals for clear visualization.

```
query = """
SELECT EXTRACT(YEAR FROM invoice_date) AS year,
       SUM(total) AS revenue
FROM invoice
GROUP BY EXTRACT(YEAR FROM invoice_date)
ORDER BY year
"""

df_trend = pd.read_sql(query, engine)
print(df_trend)
df_trend["year"] = df_trend["year"].astype(int)
```

```
     year  revenue
0  2021.0   449.46
1  2022.0   481.45
2  2023.0   469.58
3  2024.0   477.53
4  2025.0   450.58
```

INSIGHT 2 REVENUE HEALTH CHECK

```
df_trend.plot(
    x="year",
    y="revenue",
    kind="line",
    marker="o",
    figsize=(8,4),
    title="Sales Trend Over Time"
)

plt.show()
```



Outcome:

- Highly volatile sales trend.
- Peaks in 2022/2024, followed by drastic drops in 2023/2025.

Business Implication:

- Revenue relies dangerously on sporadic blockbuster album releases.

Action:

Transition to a recurring revenue model. Introduce a Monthly Subscription Tier or Loyalty Programs to stabilize cash flow during slow months

Q3. GENRE ANALYSIS POPULARITY

Mindset: What are our customers actually listening to? How should we optimize the digital catalog?

Data Direction: Need to bridge the genre name with invoice_line quantities.

Execution (SQL):

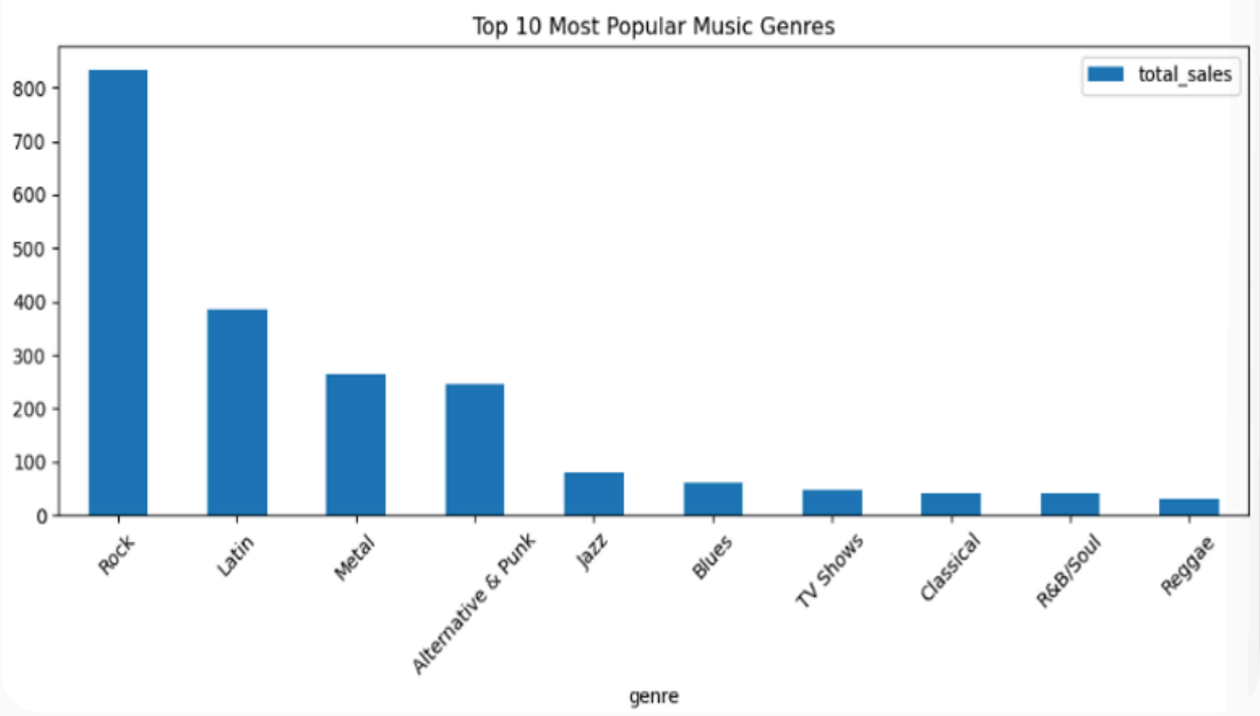
- Performed complex INNER JOIN across 3 tables (genre →track →invoice_line).
- Used GROUP BY genre name.
- Applied LIMIT 10 to filter out low-performing categories and focus on core data (removing noise).

```
query = """
SELECT g.name AS genre, SUM(il.quantity) AS total_sales
FROM genre g
JOIN track t ON g.genre_id = t.genre_id
JOIN invoice_line il ON t.track_id = il.track_id
GROUP BY g.name
ORDER BY total_sales DESC
LIMIT 10
"""

df_genre = pd.read_sql(query, engine)
print(df_genre)
```

	genre	total_sales
0	Rock	835
1	Latin	386
2	Metal	264
3	Alternative & Punk	244
4	Jazz	80
5	Blues	61
6	TV Shows	47
7	Classical	41
8	R&B/Soul	41
9	Reggae	30

INSIGHT 3 PRODUCT PORTFOLIO OPTIMIZATION

```
df_genre.plot(
    x="genre",
    y="total_sales",
    kind="bar",
    figsize=(10,5),
    title="Most Popular Music Genres"
)

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Outcome:

- **"Rock"** is overwhelmingly the **#1 genre** (835 tracks sold).
- Sells more than double the 2nd place "Latin" (386 tracks).

Business Implication:

- Chinook's core demographic consists mainly of Rock/Metal fans.

Action: The Licensing Team must prioritize acquiring new/classic Rock tracks. Email marketing campaigns should heavily feature Rock releases for higher conversion rates.

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EXECUTIVE SUMMARY

Geographic Dominance:

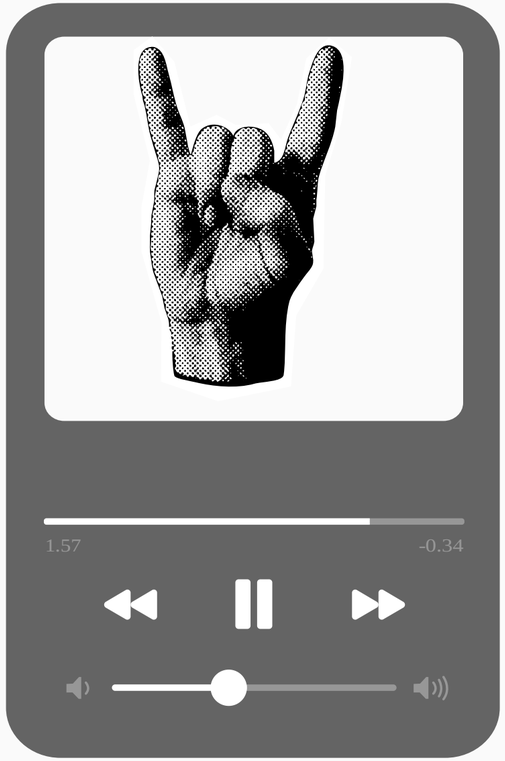
- North America (USA & Canada) is the absolute profit center, far exceeding European markets.

Revenue Volatility:

- Sales are highly unpredictable. The company relies too heavily on sporadic blockbuster album drops rather than consistent buying habits.

Product Preferences:

- "Rock" is the foundational pillar of Chinook's digital catalog, outperforming all other genres combined.



STRATEGIC RECOMMENDATIONS

1. Revenue Stabilization (Subscription Model):

- Shift from a purely transaction-based model to recurring revenue.
- Introduce a "**Chinook VIP**" monthly subscription or loyalty points program to maintain steady cash flow during slow months.

2. Focused Catalog Expansion (Licensing):

- Allocate the majority of the purchasing budget to acquire classic and trending Rock & Metal albums.
- Satisfy the core demographic to maximize ROI.

3. Geo-Targeted Marketing & Cross-Selling:

- Concentrate ad spend and VIP care on the USA and Canada.
- Use Bundle Offers (e.g., Buy Rock, get a discount on Jazz/Blues) to clear digital inventory and encourage cross-genre listening.

THANK YOU